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The drone that will never crash

DJI, Epson, and Y Media Labs have teamed up to create an AR drone flight simulator app for the Epson Moverio BT-300 smart glasses <http://digit.in/DronesAR>

In 1896, Sir Edward Henry, Inspector General of the Bengal Police, who was said to be looking for a replacement for the failing anthropometries as a method of identifying criminals, consulted Sir Francis Galton. After its implementation, one of Henry's workers, Azizul Haque, created a process to classify and store the prints in easy and accurate searches. The Henry Classification System, as it came to be known, was gradually adopted around the world and is said to be the precursor to the modern fingerprinting systems.

LOOKING GOOD

The credit as the father(s) of modern face recognition goes to Woody Bledsoe, Helen Chan Wolf, and Charles Bisson. Their manual face recognition system was capable, with moderate accuracy, of recognising a matching photo after it was provided with an input. However, a downside to this approach was that markers had to be defined manually. The next breakthrough came in the late 1980s with Eigen faces, a method that applied linear algebra to face recognition by converting images to their low-dimensional, featureless representations. This even led to the first automated facial recognition systems.

Most of the next decade in facial recognition was based on refining existing systems with larger data sets and better resolution cameras. But still, the existing systems weren't good enough for recognizing faces in crowded situations. That changed, along with other improvements, through research at some of the world's leading universities in the US and the rest of the world. Also, thanks to the adoption of newer features such as 3D face scanning, error rates have consistently dropped by half every two years. Today, face recognition along with

DID YOU Know

Alphonse Bertillon has been referenced a couple of times in Sherlock Holmes. In *The Hound of the Baskervilles*, a client refers to Holmes as the "second highest expert in Europe" after Bertillon.

In another story, the Naval Treaty, Holmes himself expresses his admiration for the Bertillon system of anthropometrics

3D face scanning, is also easily accessible through consumer gadgets, such as the iPhone X that comes with the FaceID feature and the Windows Hello feature on many Microsoft and Windows based devices.

AN EYE FOR AN EYE

While the development of iris recognition technology in the field of biometrics is fairly recent, the idea that the iris could be used for recognition dates as far back as the 1930s. The idea is said to have been first proposed by Frank Burch, an eye specialist, in 1939. But it was John Daugman's patents in 1994 that still form the basis of most modern day iris recognition systems. The patent also included computer vision algorithms that form the basis for image processing, feature extraction, and matching.

While iris recognition is said to be making its way into modern consumer devices, one of the most recent implementations on the Samsung Galaxy S8 and S8+ was reportedly beaten by something as elementary as a sufficiently high resolution image of the user's face. The Government of India, on the other hand, is also said to be forming one of the largest databases of biometric data, that includes iris scans, in the form of its Aadhaar citizen identity program.

BEHAVE YOURSELF

While most authentication mechanisms, many of which we've mentioned in the article, measure a part of the human body, there is a new and emerging field that involves measuring human behaviour. Behavioural biometrics, or biometrics involves the analysis of human behaviour as an authentication mechanism. Sounds confusing? You are probably already quite familiar with

voice-recognition based authentication systems which form the most prominent example of biometrics today. Each of us spends a significant amount of time every day with our gadgets. And as it happens, we use these gadgets in a particularly unique way, which can be effectively analysed and recognised. Using machine learning to execute pattern recognition, devices can now lock out fraudulent users if they detect deviant behaviour. For a workstation, a combination of keystroke dynamics, mouse usage, GUI interactions and behavioural patterns can form something akin to a virtual fingerprint.

IN A HEARTBEAT

In the future, we might see the field of biometrics get even more invasive than what it is today. Researchers at the University of Buffalo are said to have conceptualized a 'cardiac' scanner that will be able to recognise a person by the shape, size and beating of his heart from as far as 500m.



Image by: Biswarup Ganguly

Aadhaar is one of the largest biometric databases, and it includes iris scans.

Current technology is also expected to get more precise and convenient in the future. You can expect to see fingerprint scanners in which one has to simply wave her hand over it, become quite commonplace. Seeing such drastic strides in technology and especially in the field of biometrics in the near future, an important question now arises – would it be okay for huge corporations and government agencies to record and store information about you that is as personal as even the beating of your heart?